A Computational Framework for Analyzing *In Vivo* Joint Fluid Motion

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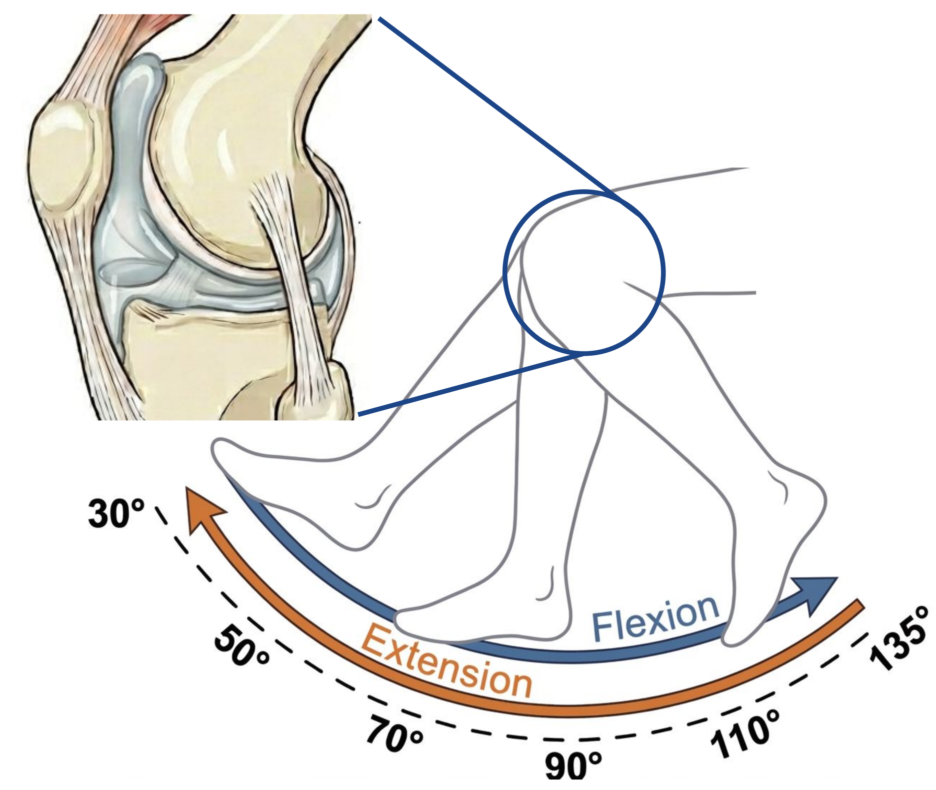
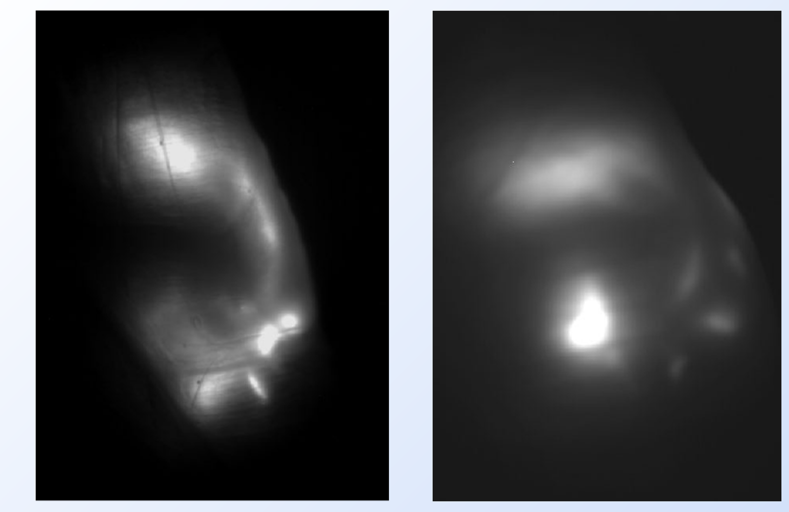
Research Objective: Develop a framework to quantitatively analyze *in vivo* synovial fluid movement in NIR fluorescence imaging video data

- **Synovial fluid (SF)** is the “lifeblood” of joints, providing nutrients and biomechanical protection throughout life
- **Near-infrared (NIR) fluorescence imaging** is a powerful tool to non-invasively visualize biological processes *in vivo*
- **Biomedical image analysis** is notoriously difficult: uneven lighting, soft boundaries, and large specimen variation – requiring much customization
- **Biomedical video analysis** introduces more challenges: camera shake, information density, and generalization across frames





Computational framework must also handle anatomical variance *between specimens*

Methodology

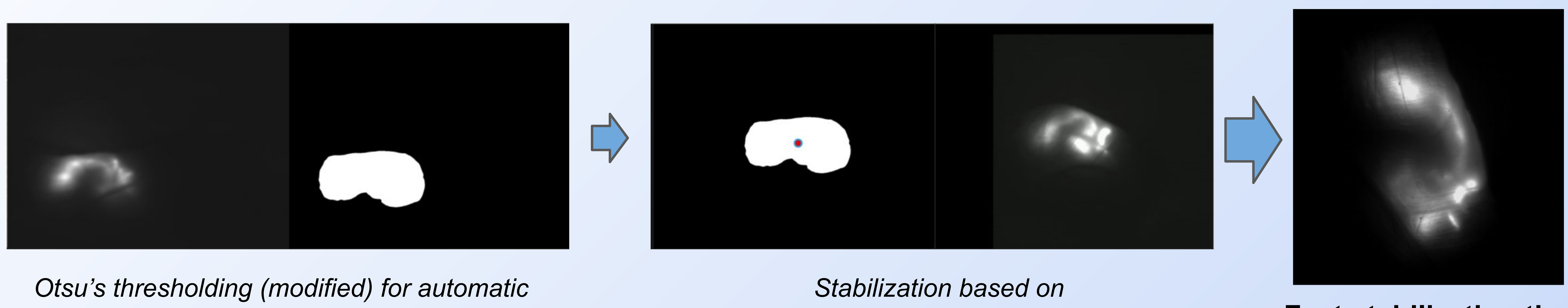
1. Stabilization

Key Challenges:

- Classical methods fail due to blurry boundaries
- Deep learning inapplicable due to lack of labeled data and pre-trained models

Solution:

Automatic stabilization using point estimate of fluid centre



Otsu’s thresholding (modified) for automatic outer boundary estimation

Stabilization based on estimated centre

Parallelized for scalability

Fast stabilization that generalizes *over entire dataset*

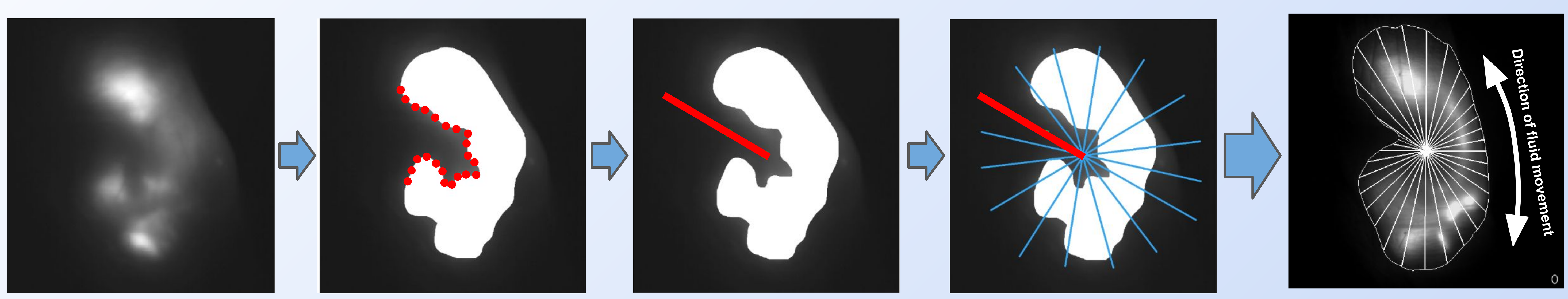
2. Signal Extraction

Key Challenges:

- Flow not directly measurable; only changes in intensity
- Knee rotates during motion, despite being centered

Solution:

Discretization relative to invariant reference point



Detailed thresholding for inner boundary estimation

Estimated femur position yields reference point for fluid

***Radial segmentation* for signal extraction**

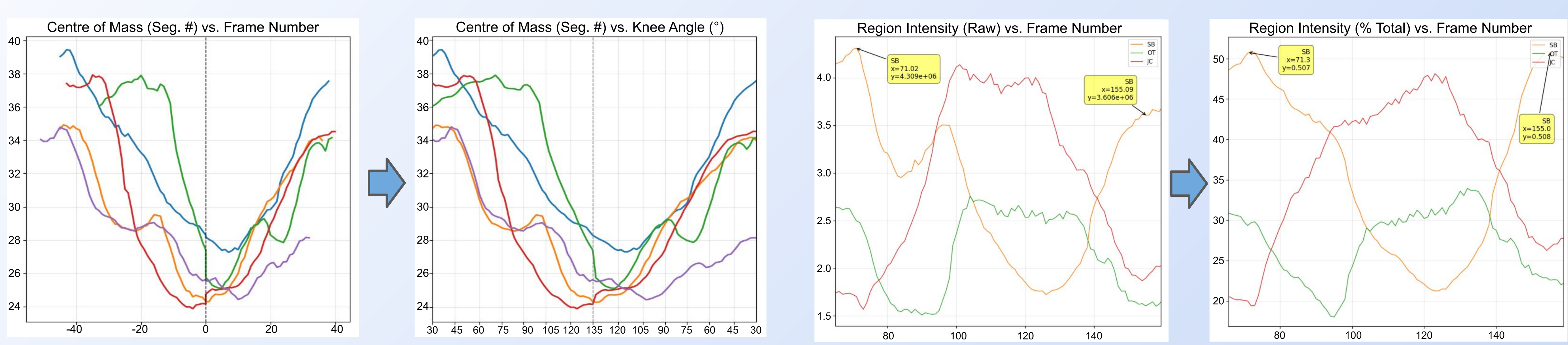
3. Normalization

Key Challenges:

- Slightly different duration for each joint motion cycle
- Significant brightness variation between videos

Solution:

Temporal and intensity rescaling for quantity alignment



Convert time axis to angle of joint flexion

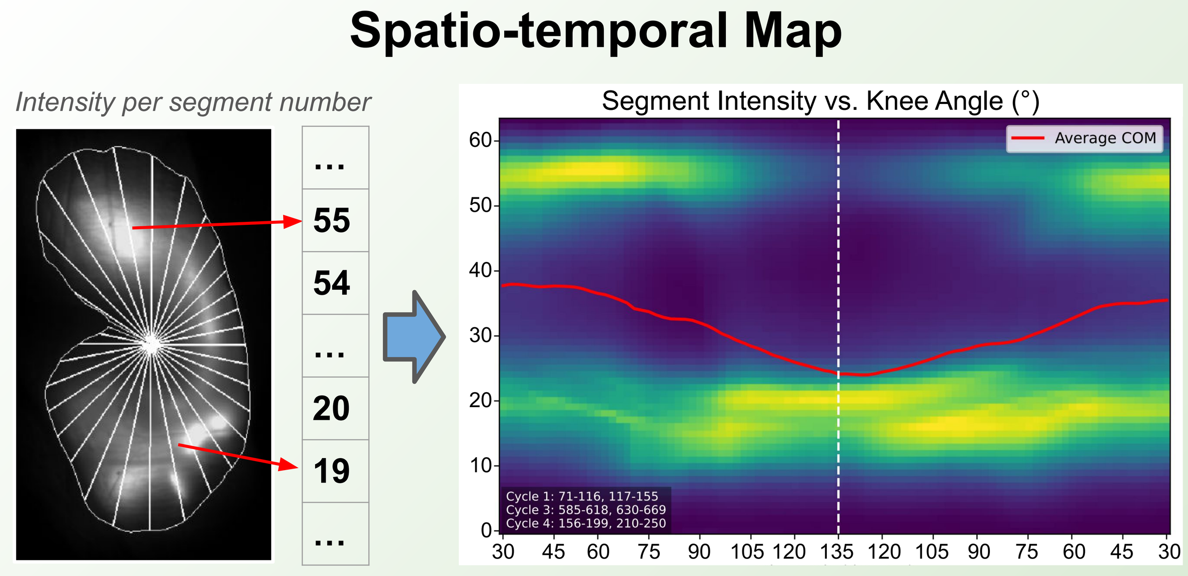
Convert intensity measurements to % of total

Results

Overview

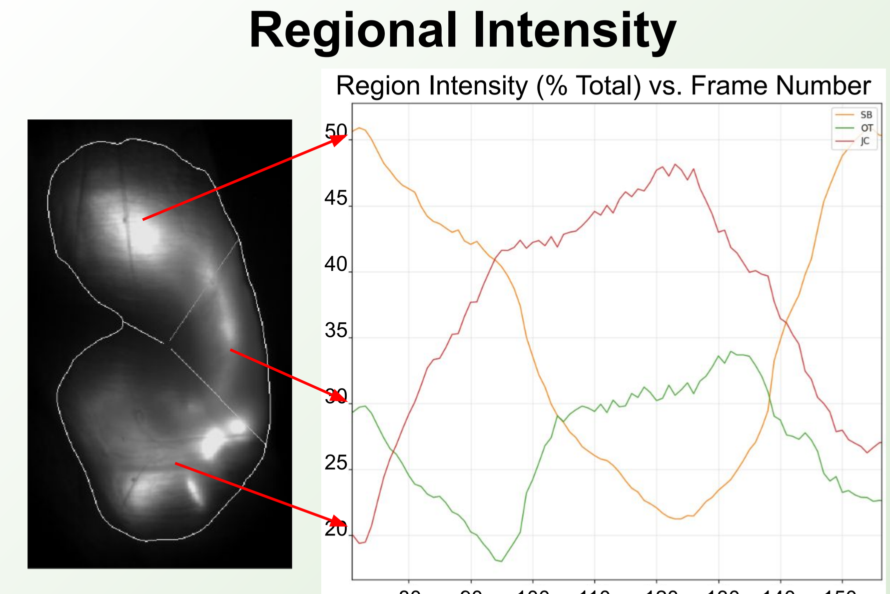
- Rich visualization based on unique *radial segmentation*
- Scalable over large amounts of video data (>3000 frames per hour)

Spatio-temporal Map



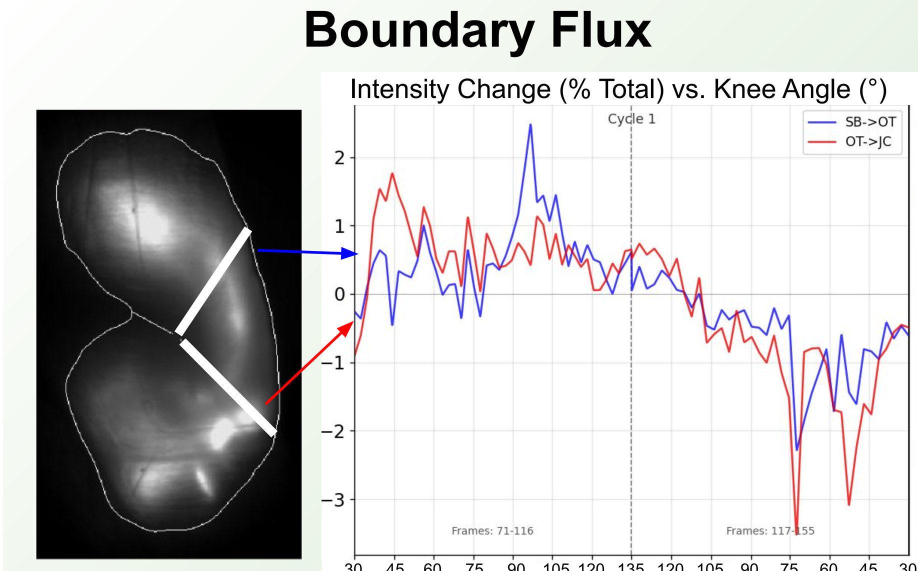
Snapshot of dye *distribution through time*, enabling comparison

Regional Intensity



Dye quantity within anatomical regions over time

Boundary Flux



Dye transport rate between anatomical regions

Conclusion

- **Signal extraction and quantitative analysis** pipeline for real *in vivo* video
- **Over 26x speedup** compared to manual methods – analysis completed in *hours*, not weeks
- **Transparent, deterministic, open-source** (see QR code) for inspection and reproducibility

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